FUNCTIONS IN MACHINE LEARNING

- Disease Identification from X-rays: *f(X-ray image) = disease diagnosis*
- Treatment Recommendations: *f(patient symptoms and history) = suggested treatments*
- Epidemic Outbreak Prediction: *f(recent health data) = risk of epidemic outbreak*
- Predictive Health Monitoring: *f(real-time health metrics) = immediate health risks*
- Clinical Note Analysis: *f(clinical notes) = disease diagnosis or treatment recommendation*
- Drug Side Effect Identification: *f(patient feedback) = potential drug side-effects*
- Epidemic Surveillance from News: *f(news articles) = potential epidemic outbreaks*
- Brain MRI Analysis: *f(MRI image) = presence of tumors or anomalies*
- Cell Image Classification in Pathology: *f(Cell image) = cell type and potential abnormalities*

SUPERVISED LEARNING AND FUNCTIONS!

In supervised learning, we have a dataset consisting of input-output pairs, usually denoted as $(x_1, y_1), (x_2, y_2), \ldots (x_n, y_n)$, where x is the input and y is the corresponding output. The goal is to learn a function f such that for a new input x_{new}, the function can predict the output y_{new} with accuracy ($y = f(x)$).

The idea of supervised learning is to learn a function f based on the training data and then test its performance on previously unseen data.